Question Bank-717

B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING

FIRST YEAR SECOND SEMESTER EXAMINATION, 2019

(According to the New Syllabus)

CSE-510221 (Digital Systems Design)

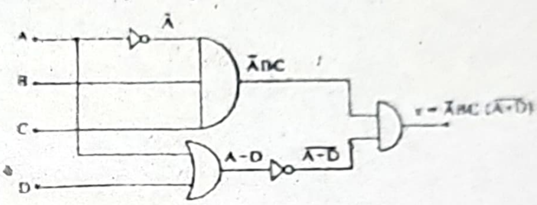
Time 3 hours

Full marks-80

[N.B. The figures in the right margin indicate full marks. Answer any four questions. Different parts of questions must be answered sequentially.]

1. (a) Define digital system. Write down importance of digital system.
   (b) Discuss the major parts of a digital computer with required diagram.
   (c) Define ASCII and Gray Code. Convert $(123.45)_8 = (?)_2 = (?)_{Hex} = (?)_{BCD}$
   (d) Write down Demorgan's theorem. Prove Demogan's theorem with 3 variables.

2. (a) What a universal gate? Prove the universality of NOR gate.
   (b) Define K-map. Using K-map simplify the expression $Y = \bar{C}(\bar{A}\bar{B}\bar{D}+D) + A\bar{B}C+\bar{D}$, also draw the logic diagram of simplified Y.
   (c) What is don't care condition of K-map? Explain.



(d) Consider the following Boolean logic diagram. If you want to get the output value x = 0 and x = 1, then what will be the input values of A, B, C, D:

4 (a) Define flipflop. Draw the logic circuit of an S-R flipflop with NAND latch and explain its logic operation
   (b) Design and discuss the MOD-6 counter.
   (c) Write down the truth table for a full adder. Draw the logic circuit and explain its operation

4. (a) Explain the operation of successive approximation ADC with required diagram.
   (b) What is the largest value of output voltage from an eight-bit. DAC that produces 1-OV for a digital input of 00101000?
   (c) Differentiate between EPROM and DRAM.
   (d) Mention the advantageous of successive approximation ADC 3 over digital Ramp ADC.

5. (a) What is encoder? Explain the working principle of 8 line to 3-line encoder with logic diagram and truth table.
   (b) Define MUX and DMUX. Why MUX is called data selector? and explain it.
   (c) Draw and explain BCD to 7 segment decoders.

6 (a) What are the differences between combinational logic circuit and sequential logic circuit?
   (b) Define memory. Discuss the basic organization of a memory unit of 3 x 4 bits.
   (c) How many 32k x 8 RAM chips are needed to provide a memory capacity of 256k bytes?
   (d) Describe the functional parts of an ALU.